

Introduction to Probability and Statistics

TRAINING QUIZ #2

Q 1 – Let X be a discrete random variable defined on $V \subset \mathbb{Z}$ with a distribution $P(X = x | \theta_1, \theta_2)$ depending on two unknown parameters θ_1 and θ_2 such that $\theta_2 > \theta_1$.

1. Give the expression of the q^{th} theoretical moment :

$$E(X^q) =$$

2. Let $X_1, \dots, X_n$ be n r.v. from the distribution above.

Give the expression of the q^{th} empirical moment :

$$m_q =$$

3. Suppose that $E(X) = \theta_1 - \theta_2$ and $E(X^2) = \theta_1 + \theta_2$. Suggest a method to estimate θ_1 and θ_2 and give the estimators for each parameter.

Q 2 – Let $X_1, \dots, X_n$ be n i.i.d. r.v's with an exponential distribution $f_X(x | \theta) = 1/\theta e^{-x/\theta}$, $x \in \mathbb{R}^+$, $\theta > 0$. We know that $E(X_i) = \theta$ and $Var(X_i) = \theta^2$, $i = 1, \dots, n$.

1. Using the method of moments, find $\hat{\theta}$ an estimator of θ ?

2. An estimator $g_n(\underline{X})$ of a parameter θ is said to be consistent if it converges in probability to θ that is to say iff $\forall \varepsilon > 0$, $\lim_{n \rightarrow +\infty} P(|g_n(\underline{X}) - \theta| > \varepsilon) = 0$.

Using the Chebyshev's Inequality : $\forall \varepsilon > 0$, $P(|Y - EY| \geq \varepsilon) \leq \frac{Var(Y)}{\varepsilon^2}$.

Prove that $\hat{\theta}$ is a consistent estimator for θ .

Q 3 – Let $X_1, \dots, X_n$ be n Binomial i.i.d. r.v's with parameter (N, p) ($0 < p < 1$).

$$P(X_i = x_i | p) = C_N^{x_i} p^{x_i} (1 - p)^{(N - x_i)}, \quad x_i = 0, 1, \dots, n.$$

We want to find $\hat{p}$, the m.l.e. of p .

1. Write the likelihood

$$L(p) =$$

2. Calculate the log-likelihood

$$\log L(p) =$$

3. Give and solve the likelihood equation to express $\hat{p}$ the m.l.e. :

4. Is this estimator unbiased?